Fatty acids oxidation

Among the different foodstuffs, lipids give (through fatty acids oxidation) the maximum amount of energy. 3 Different pathways for fatty acid oxidation are present: α, β and ω.

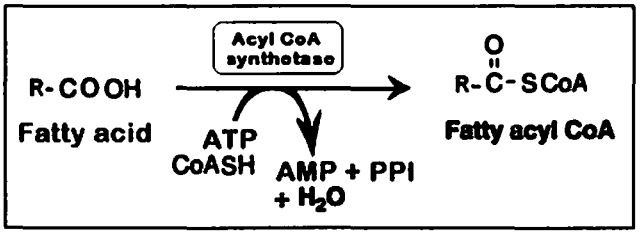
I. *β-Oxidation:*

A. Site:

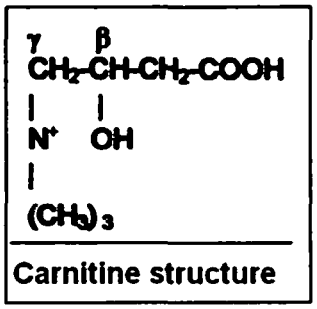
1. **Intracellular location:** Mitochondria.
2. **Organ location:**
   a) Liver, kidney, heart and skletal muscles.
   b) β-Oxidation never occur in brain and RBCs.

B. Transport of fatty acids into the mitochondria:

1. After the fatty acid is taken up by the cell, it is converted to the acyl CoA in the cytosol:



2. Because the β-oxidation occurs in the mitochondrial matrix, the acyl CoA must be transported across the mitochondrial inner membrane, which is impermeable to acyl CoA. Therefore, a specialized carrier present in the inner mitochondrial membrane called **carnitine**, transports the acyl CoA from cytosol into the mitochondrial matrix. The transport process is called carnitine shuttle.
3. **Carnitine shuttle:**
   a) **Structure of carnitine:**
   Carnitine is β-hydroxy-γ-trimethyl ammonium butyric acid. It is derived from lysine amino acid.



Carnitine structure

   b) **Function of carnitine:**
   It transports long chain acyl CoA inside the mitochondrial matrix where enzymes for β oxidation are present.
   c) **Steps of shuttle:** Three enzymes are involved:
      1) **Carnitine acyl transferase I:**
         i- For palmitic acid, it is called carnitine palmitoyl transferase I
         ii- It is present in the outer mitochondrial membrane.

iii- It transfers the acyl group from acyl CoA to carnitine to form acyl carnitine.

2) **Carnitine acylcarnitine translocase:**

i- It is present in the inner mitochondrial membrane.

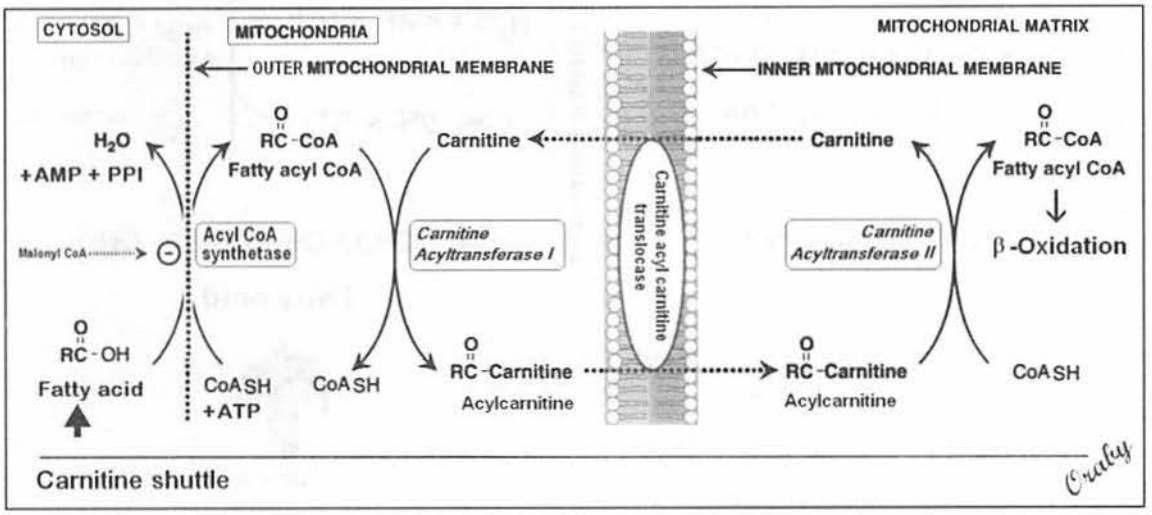
ii- It transports acyl carnitine across inner mitochondrial membrane (in exchange with carnitine).

3) **Carnitine acylcarnitine transferase II:**

i- For palmitic acid, it is called carnitine palmitoyl transferase II.

ii- It is present in the inner mitochondrial membrane.

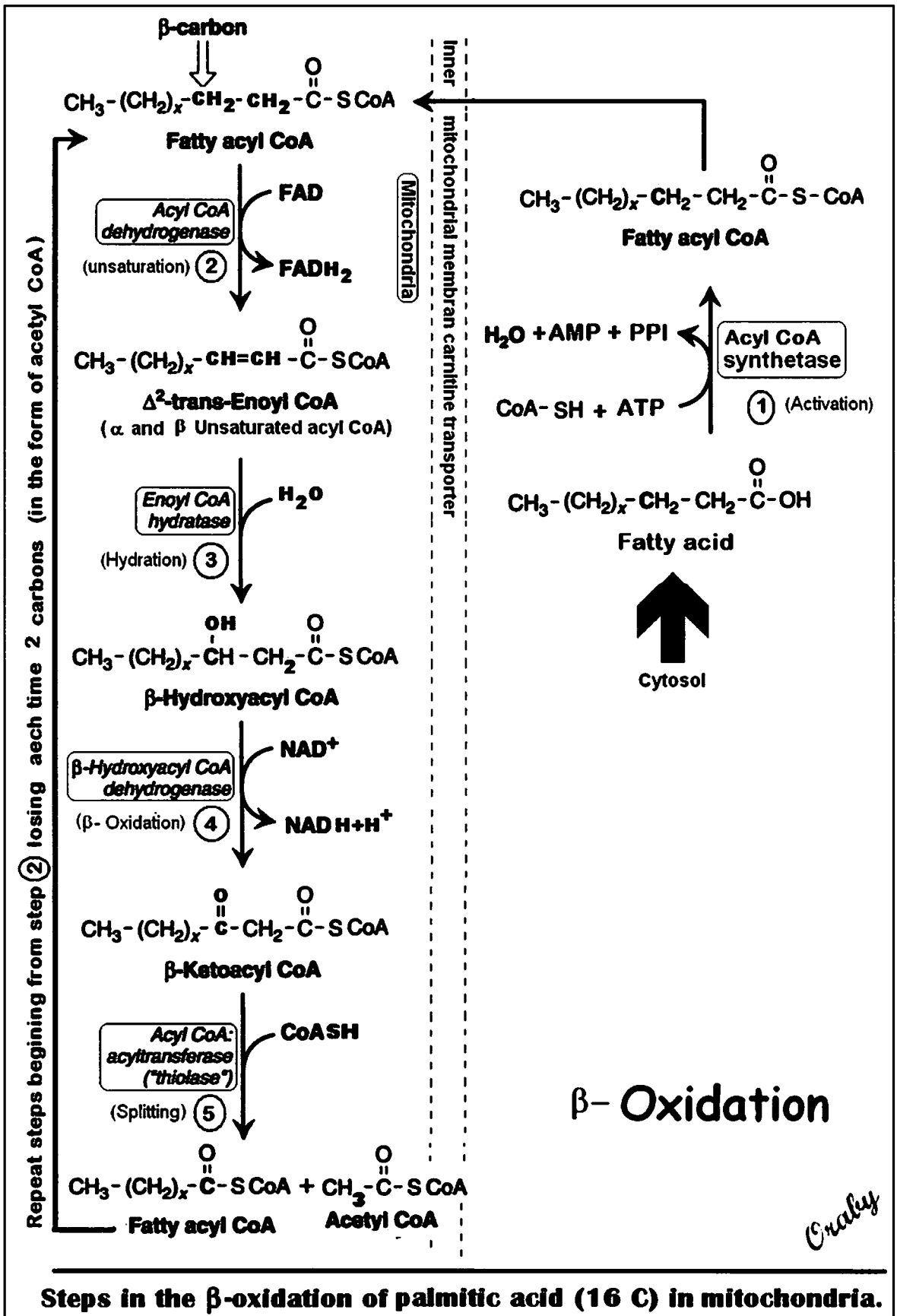
iii- It transfers the acyl group from acyl carnitine to form acyl CoA again.



C. Steps of β-Oxidation:

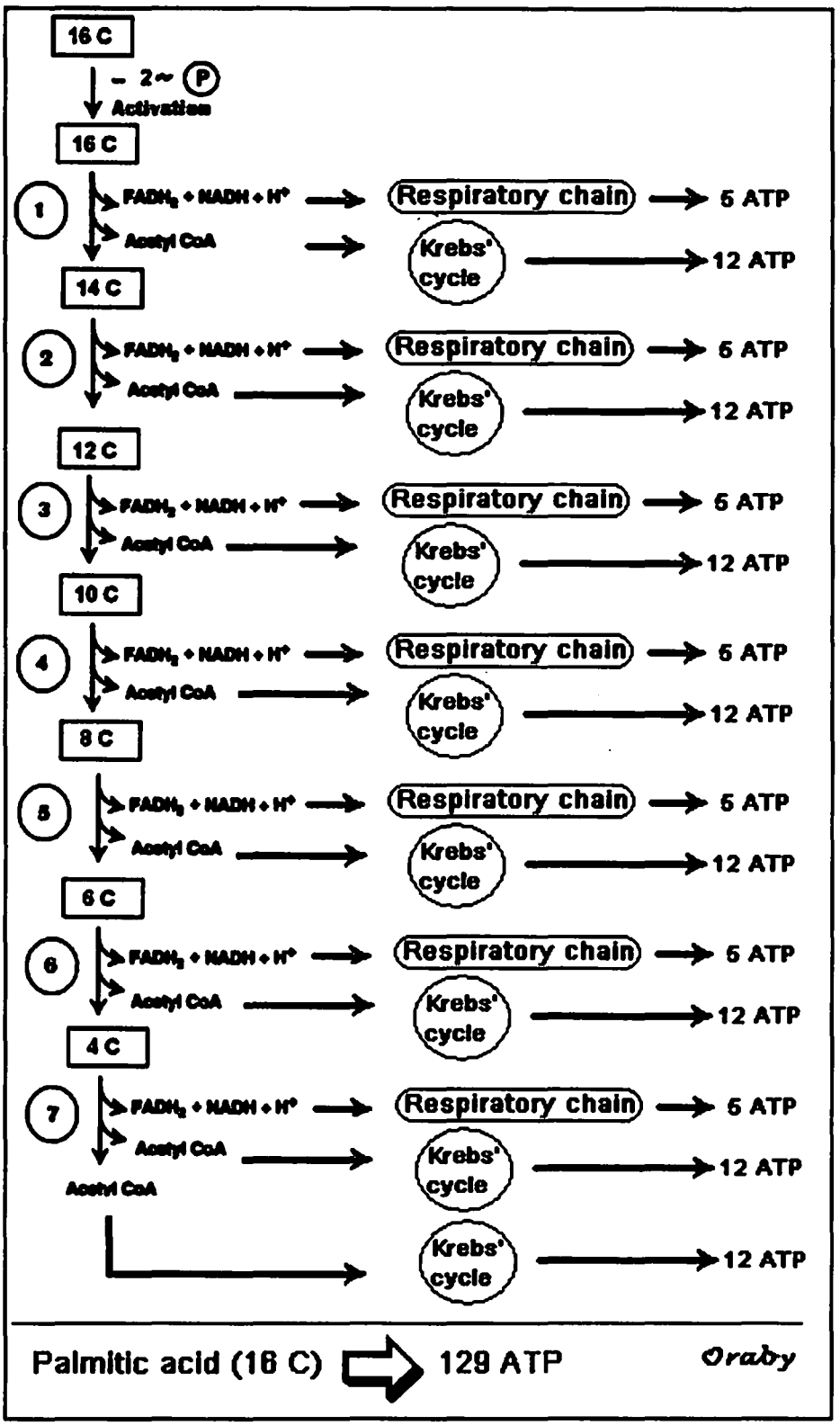
1. **Activation of fatty acid and transport inside the mitochondria:**
   - *a)* Catalyzed by fatty acyl CoA synthetase enzyme and carnitine shuttle.
   - *b)* 2 high energy bonds (~P) are utilized: ATP ➔ AMP + PPi.
2. **Unsaturation of fatty acids:**
   - *a)* Catalyzed by fatty acyl CoA dehydrogenase enzyme.
   - *b)* This enzyme is one of flavoproteins and its reduced coenzyme ($FADH_2$) gives 2 ATP when oxidized in the respiratory chain.
3. **Hydration:**
   - *a)* Catalyzed by enoyl CoA hydratase enzyme.
   - *b)* It helps the addition of water to saturate double bonds.
4. **Oxidation:**
   - *a)* Catalyzed by β-hydroxyacyl CoA dehydrogenase enzyme.
   - *b)* Its reduced coenzyme ($NADH+H^+$) gives 3 ATP when oxidized in the respiratory chain.
5. **Splitting (cleavage) step:**
   - *a)* Catalyzed by Acy CoA acyl transferase (thiolase) enzyme.

b) It splits acyl CoA into acetyl CoA and acyl CoA (shorter than the original one by 2 carbon atoms).



Steps in the β-oxidation of palmitic acid (16 C) in mitochondria.

D. Energy production of β-oxidation:



1. e.g. palmitic acid (16 carbons):
   a) β-oxidation of palmitic acid will be repeated 7 times (turns) to produce 8 acetyl CoA.

b) In each turn, one molecule of reduced $FADH_2$ and one molecule of reduced $NADH+H^+$ are produced. They are oxidized in respiratory chain to give 5 ATP.

$FADH_2$ ➔ 2 ATP

$NADH+H^+$ ➔ 3 ATP

∴7 turns × 5 ATP → 35 ATP.

c) Oxidation of one molecule of acetyl CoA in Krebs' cycle gives 12 ATP.

∴8 Acetyl CoA × 12 ATP = 96 ATP

d) Two high energy phosphate bonds are utilized in the first reaction (catalyzed by acyl CoA synthetase) which occurs for one time only.

e) **∴ Net energy gain =**

Energy produced – Energy utilized

=(35 ATP + 96 ATP) – 2 ATP

=131 ATP – 2 ATP = 129 ATP

2. **Calculation formula of energy production of oxidation of any fatty acid:**

$$= \left[\left(\frac{N}{2}-1\right)x\,5\,ATP\right]+\left[\frac{N}{2}x\,12\,ATP\right]-2\,ATP$$

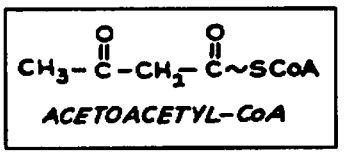
Where N = Number of carbons of fatty acid e.g. palmitic acid = 16 carbons, so energy production =

$$= \left[\left(\frac{16}{2}-1\right)x\,5\,ATP\right]+\left[\frac{16}{2}x\,12\,ATP\right]-2\,ATP$$

={(8 – 1) × 5 ATP} + {8 x 12 ATP} – 2 ATP = 129 ATP

E. Importance (functions) of β-oxidation:

1. **Energy production** e.g. palmitic acid produces 129 ATP.
2. **Production of acetyl CoA** which enter in many pathways (see sources and fate of acetyl CoA).
3. **Ketone bodies formation**: Acetoacetyl CoA is the last 4 carbons product in the course of β-oxidation of even numbered fatty acids. It may be converted into acetoacetate; one of ketone bodies (see later).



ACETOACETYL-CoA

F. β-oxidation of odd number fatty acids:

1. They are oxidized by β-oxidation till a propionyl CoA (3 carbons) is produced. Then propionyl CoA is converted to succinyl CoA as shown in the following figure.
2. **Sources and Fate of succinyl CoA:**

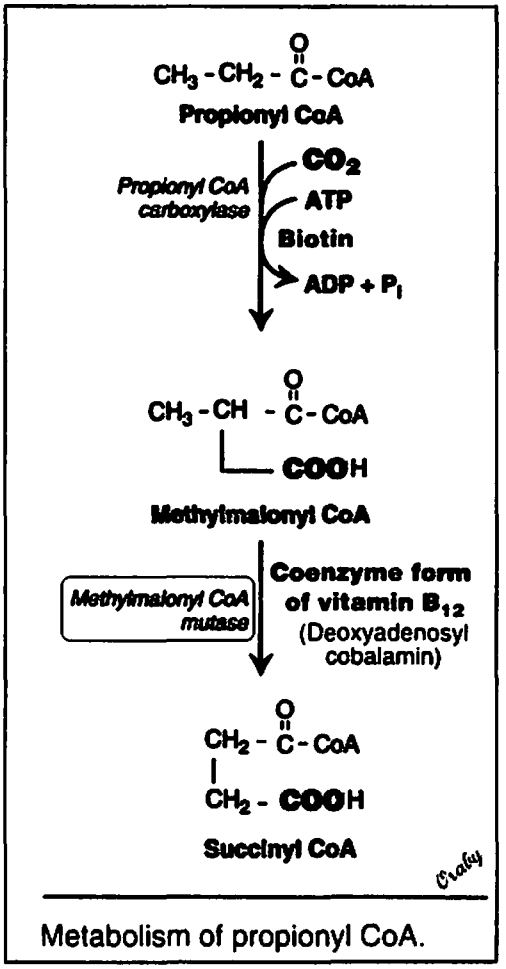
a) **Sources:**

1) Oxidation of odd number fatty acids.

2) Citric acid cycle.
3) Catabolism of some amino acids e.g. leucine, valine and methionine.

b) **Fate:**

1) Glucose synthesis (gluconeogenesis).
2) Heme synthesis.
3) Oxidation in citric acid cycle.
4) Activation of ketone bodies.
5) Detoxication.



Metabolism of propionyl CoA.

II. *α-Oxidation:*

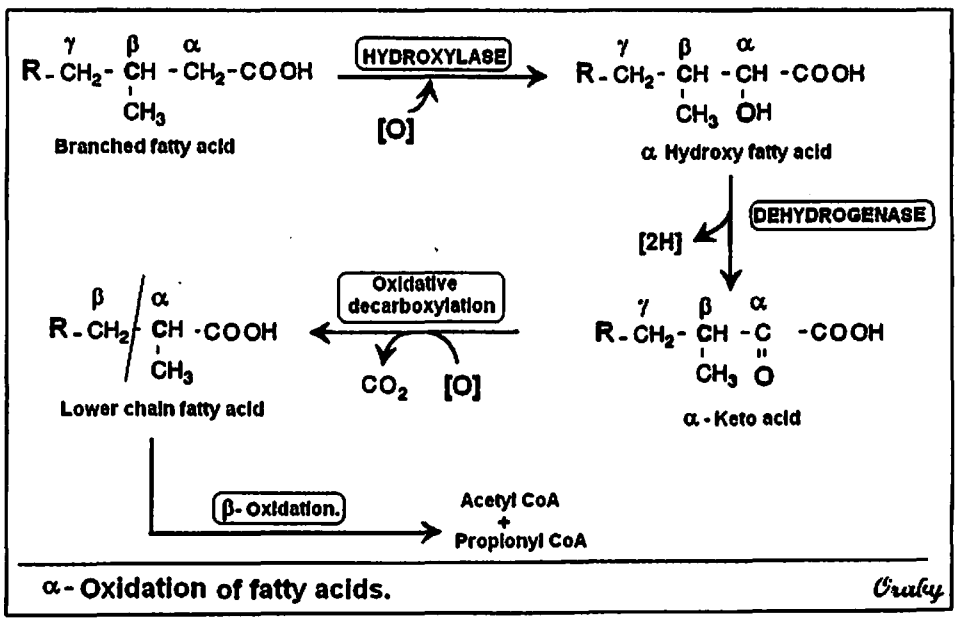
A. <u>This type of oxidation occurs in α position and characterized by:</u>

1. It is a mechanism mainly for oxidation of branched chain fatty acid, which are methylated at β position.
2. It is specific for oxidation of phytanic acid (3, 7, 11, 15 tetramethyl palmitic acid), present in plant foodstuffs.
3. It is minor pathway for fatty acid oxidation, occurs mainly in brain and nervous tissues.

B. In α-oxidation, there is one carbon atom removed at a time from α position.

C. It does not require CoASH and does not generate high energy phosphate.

D. <u>Steps:</u>



α-Oxidation of fatty acids.

E. **Refsum's disease:**

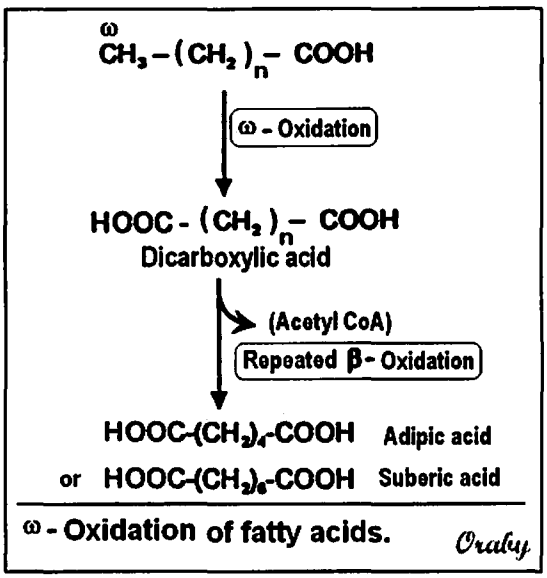
1. **This is inherited deficiency of enzymes responsible for** α oxidation of phytanic acid. This leads to accumulation of phytanic acid in nervous tissue and produce nervous damage e.g. deafness and blindness.

III. ω-Oxidation:

A. It is oxidation of **terminal CH_3** group (ω carbon) of fatty acid.

B. This produces dicarboxylic fatty acids. By β oxidation they are converted to adipic acid (6 carbons) and suberic acid (8 carbons).

C. ω-Oxidation is a minor pathway for fatty acid oxidation and catalyzed by hydroxylase enzymes of cytochrome P450.



SOURCES AND FATE OF ACTIVE ACETATE (ACETYL CoA)

I. Sources:

A. **Lipids**: Oxidation of fatty acids and ketone bodies.

B. **Carbohydrate**: Glucose oxidation ➔ Pyruvate ➔ Acetyl CoA.

C. **Proteins**:

1. **Ketogenic amino acids**: Give directly acetyl CoA or indirectly give acetoacetate ➔ Acetyl CoA.
2. **Glucogenic amino acids** ➔ Pyruvate ➔ Acetyl CoA.

II. Fate:

A. **Oxidation**: Through Krebs' (citric acid cycle).

B. **Lipogenesis**: Formation of fatty acids and triacylglycerols.

C. **Ketogenesis**: Synthesis of ketone bodies.

D. **Acetylcholine synthesis.**

E. **Cholesterol synthesis**: which is the precursor for:

1. Bile acids.
2. Vitamin D_3.
3. Steroid hormones: glucocorticoids, mineralocorticoids, male sex hormones (testosterone) and female sex hormones (estrogens and progesterone).